25901659 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCECT101 - PYTHON FOR ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer any **35** questions

Each question carries **1** marks

1. What is the output of 10 ^ 6? [U] / [CO1]

   a). 12 b). 4 c). 6 d). 8

2. Which company developed the BeagleBoard? [K] / [CO1]

   a). Raspberry Pi Foundation b). Texas Instruments

   c). Arduino LLC d). Intel

3. What happens if the slice indices are out of range? [U] / [CO1]

a). Returns available part b). Error

c). Returns None d). Returns full string

4. What will the following code output? `print(10, 20, 30, sep=", ")` [U] / [CO1]

a). 10, 20, 30 b). 10 20 30 c). 102030 d). Error

5. What is the output of not (False or not (True and False))? [U] / [CO1]

a). TRUE() b). FALSE() c). None d). Error

6. Which of the following is NOT a valid variable name in Python? [K] / [CO1]

a). _var b). var_ c). var-1 d). var1

7. What will list(range(2, 10, 2)) return? [K] / [CO1]

a). [2, 4, 6, 8] b). [2, 4, 6, 8, 10]

c). [2, 3, 4, 5, 6, 7, 8, 9] d). [2, 6, 10]

8. Which filesystem is recommended for Raspberry Pi's microSD card? [U] / [CO1]

a). NTFS b). FAT32 c). ext4 d). exFAT

9. What is Python? [K] / [CO1]

a). A web browser b). A high-level programming language

c). A type of database d). A markup language

10. What is the output of False or False? [U] / [CO1]

a). TRUE() b). FALSE() c). None d). Error

11. Which port is used to connect a camera module? [K] / [CO1]

a). HDMI b). USB c). CSI d). GPIO

12. What is the correct syntax to convert a list to a tuple? [K] / [CO1]

a). tuple(list) b). list(tuple) c). convert(list) d). tuple.toList()

13. What is the output of 5 > "5"? [U] / [CO1]

a). TRUE() b). FALSE() c). None d). Error

14. What is the output of 2 ** 4? [K] / [CO1]

a). 16 b). 8 c). 6 d). 24

15. What is the output of 10 ** (2 + 3) * 4? [U] / [CO1]

a). 400000 b). 100000 c). 40000 d). 10000

16. Which Raspberry Pi model is most suitable for AI and machine learning applications? [U] / [CO1]

a). Raspberry Pi Zero 2 W b). Raspberry Pi 3 Model B+

c). Raspberry Pi 4 d). Raspberry Pi Compute Module

17. What is the output of 5 <= 3? [K] / [CO1]

a). TRUE() b). FALSE() c). None d). Error

18. What is the result of the following code? `type(100)` [K] / [CO1]

a). int b). flaot

c). list d). string

19. What does `x <<= 3` do? [K] / [CO1]

a). Left shifts x by 3 bits and assigns b). Right shifts x by 3 bits and assigns

c). Multiplies x by 3 and assigns d). Divides x by 3 and assigns

20. Which of the following statements is true about lists in Python? [K] / [CO1]

a). Lists are immutable

b). Lists are ordered and mutable

c). Lists store only integers

d). Lists do not allow duplicates

21. Which of the following is an alternative OS for Raspberry Pi? [K] / [CO1]

a). Windows 10 b). Android c). MacOS d). Linux

22. What will be the output of s[::-2] if s = "abcdef"? [U] / [CO1]

a). fdb b). ec c). fb d). fd

23. What is the value of x after x = 10; x *= 2; x %= 3? [U] / [CO1]

a). 2 b). 10 c). 20 d). Error

24. What is the output of the following code? for i in range(2, 8, 3): print(i) else: print("Else block") [K] / [CO2]

a). 2, 5, Else block b). 2, 5, 8, Else block

c). Else block d). 2, 3, 4, Else block

25. What is the correct syntax for an if-elif-else statement? [U] / [CO2]

a). if-elif-else b). if-else-elif c). elif-if-else d). else-elif-if

26. Which method is used for old-style string formatting in Python? [U] / [CO2]

a). format() b). f-string c). % operator d). template()

27. What is the output of the following code? x = 1 while x < 3: print(x) x += 1 else: Print("Else block") [K] / [CO2]

a). 1, 2, Else block b). 1, 2

c). Else block d). 1, 2, 3, Else block

28. What will be the output of if 10 < 5: print("A") elif 5 >= 5: print("B") else: print("C")? [U] / [CO2]

a). A b). B c). C d). Error

29. What does 'hElLo WoRlD'.capitalize() return? [K] / [CO2]

a). 'Hello world' b). 'hElLo WoRlD' c). 'HELLO WORLD' d). 'Hello World'

30. What is the output of the following code? for i in range(0, 10, 3): print(i) [U] / [CO2]

a). 0, 3, 6, 9 b). 0, 3, 6, 9, 12

c). 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 d). 3, 6, 9

31. How many times will the loop run? x = 1 while x < 10: X *= 2 [U] / [CO2]

a). 4 b). 5 c). 3 d). 6

32. Which command is used to clean up GPIO settings in Python? [A] / [CO3]

a). GPIO.cleanup() b). GPIO.reset()

c). GPIO.clear() d). GPIO.flush()

33. How do you set the title of a Tkinter window? [A] / [CO3]

a). window.set_title("Title") b). window.title("Title")

c). window.label("Title") d). window.heading("Title")

34. Which function is used to write a digital output to a GPIO pin in RPi.GPIO? [An] / [CO3]

a). GPIO.set(pin, state) b). GPIO.write(pin, state)

c). GPIO.output(pin, state) d). GPIO.digitalWrite(pin, state)

35. Which module is used to create GUI applications in Python? [K] / [CO3]

a). pygame b). tkinter c). matplotlib d). pandas

36. Which tool can be used to check the GPIO pin layout on a Raspberry Pi? [An] / [CO3]

a). pinout b). gpio-info

c). gpio-map d). pin-map

37. What type of variable is typically used with a Checkbutton widget? [C] / [CO3]

a). IntVar b). StringVar c). BooleanVar d). FloatVar

38. Which geometry manager positions widgets at exact coordinates? [C] / [CO3]

a). place() b). pack() c). grid() d). position()

39. Which widget is used to display text in Tkinter? [A] / [CO3]

a). Label b). Button c). Entry d). Canvas

40. What function is used to configure a GPIO pin as an output? [An] / [CO3]

a). GPIO.setup(pin, GPIO.OUT) b). GPIO.output(pin, GPIO.OUT)

c). GPIO.setmode(GPIO.OUT) d). GPIO.init(pin, GPIO.OUT)

[1x35 = 35]

25901659 **B**

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4. What function is used to configure a GPIO pin as an output? [An] / [CO3]

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